

Ethan Nguyen

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EDUCATION

Drexel University

B.S. Computer Science

Philadelphia, PA

Expected June 2027

- GPA: 3.39
- Relevant Coursework: Data Structures, Software Engineering, Computer Graphics, Computer Programming, UI/UX Design

EXPERIENCE

Colossus Studios

Software Engineer Co-op

Remote (Austin, TX)

Sept 2024 – Apr 2026

- Contributed to Wavefront Platform, a GPU-accelerated system generating 3D Gaussian splat digital twins from video using computer vision pipelines
- Implemented real-time WebGPU collision detection and physics inspired by BabylonJS/Havok and NVIDIA research
- Developed angular momentum, friction, and placement correction systems for interactive 3D environments
- Performed GPU profiling, stress testing, and readback optimization to improve rendering performance
- Researched open-source reconstruction pipelines to reduce COLMAP dependency and cloud GPU cost
- Built React/TypeScript UI features including mobile layouts, file management tools, and iframe-based viewer synchronization
- Integrated Stripe payments and supported AWS/Vercel-deployed backend services
- Collaborated with mentor and interns on debugging, detection logic, and feature development

PROJECTS

Real-Time ASCII Vision System | *Python, OpenCV, NumPy, Pillow*

- Built real-time computer vision system converting webcam and images into ASCII animation streams
- Implemented grayscale threshold mapping and symbol quantization for dynamic rendering
- Generated animated GIF exports from ASCII-rendered frame sequences

Survey System | *Java, OOP*

- Developed CLI survey system supporting creation, completion, and grading workflows
- Applied object-oriented design and serialization for persistent state management

UX Research Interface Design | *Figma, User Research*

- Designed sustainability-focused interface using usability testing and iterative feedback

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C/C++, SQL

Web & Cloud: React, Node.js, AWS, Vercel, Stripe API, HTML/CSS

Graphics & Systems: WebGPU, GPU Computing, Computer Vision, Physics Simulation, OpenCV

Tools: Git, Docker (basic), Linux/WSL, Figma